Multiple-choice questions

|!|BCEMC0170000001|!|

Which of the following does ***not*** belong to the central nervous system?

A. cerebellum
B. cranial nerve
C. medulla oblongata
D. spinal cord

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000002|!|

Which of the following belong(s) to the central nervous system?

(1) medulla oblongata
(2) spinal cord
(3) spinal nerves

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000003|!| ★

Which of the following combinations about the functions of the peripheral nervous system and the central nervous system is correct?

	Peripheral nervous system	***Central nervous system***
A.	coordinates involuntary actions	coordinates voluntary actions
B.	receives sensory stimuli	integrates sensory information
C.	integrates sensory information	receives sensory stimuli
D.	produces sensory information	produces movements

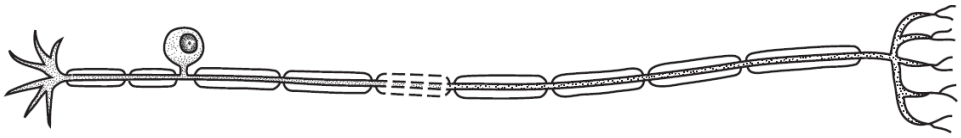
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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000004|!|

What type of neurone is shown in the diagram below?



A. sensory neurone
B. interneurone
C. motor neurone
D. cannot be identified

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000005|!| ★

The diagram below shows a sensory neurone.



Which of the following statements about the neurone is correct?

A. The neurone is connected to the brain at P.
B. Electrical impulses are transmitted from P to Q.
C. The cell body of the neurone is located in the central nervous system.
D. The neurone is synapsed to an effector at Q.

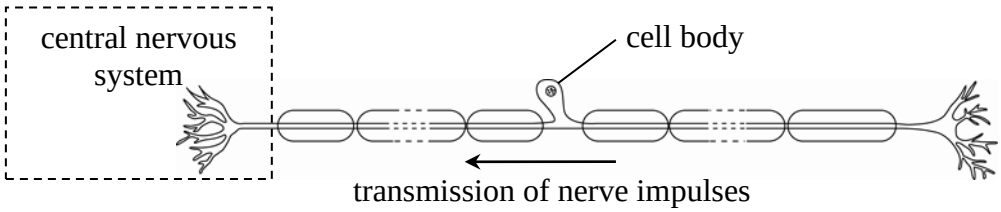
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0170000006|!| ★

The diagram below shows a neurone.



Which of the following combinations correctly identifies the neurone and gives the reason?

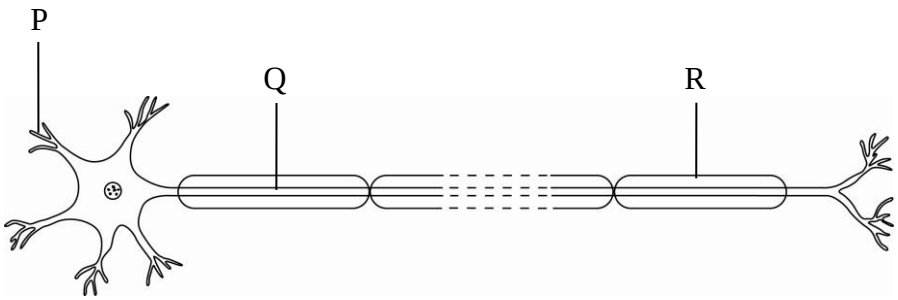
	Type of neurone	***Reason***
A.	interneurone	the axon and the dendron have the same length
B.	sensory neurone	the axon is longer than the dendron
C.	sensory neurone	the cell body is located outside the central nervous system
D.	motor neurone	the neurone transmits nerve impulses away from the central nervous system

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0170000007|!|

The diagram below shows a motor neurone.



What are structures P, Q and R?

	P	***R***	***Q***
A.	axon	dendron	myelin sheath
B.	dendron	axon	myelin sheath
C.	axon	dendron	connective tissue
D.	dendron	axon	connective tissue

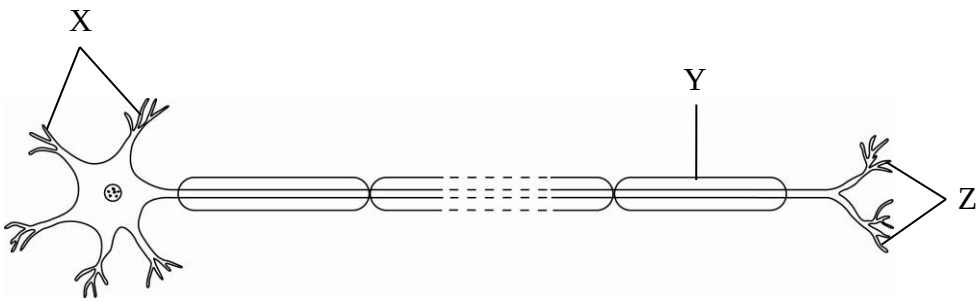
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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0170000008|!|

The diagram below shows a motor neurone.



What are structures X, Y and Z?

	X	***Y***	***Z***
A.	motor end plates	myelin sheath	dendrites
B.	motor end plates	cell body	dendrites
C.	dendrites	myelin sheath	motor end plates
D.	dendrites	cell body	motor end plates

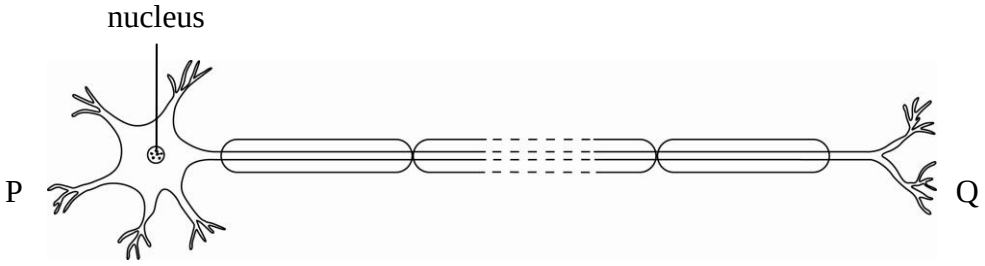
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0170000009|!| ★★

The diagram below shows a motor neurone.



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What structures could be found at P and Q ends?

	P	***Q***
A.	brain	retina of the eye
B.	spinal cord	leg muscle
C.	retina of the eye	brain
D.	leg muscle	spinal cord

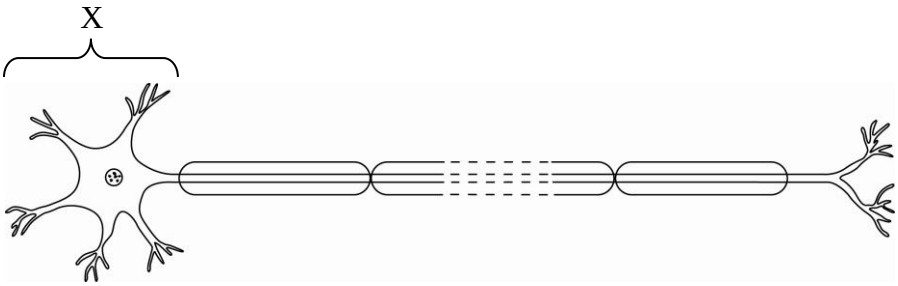
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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000010|!| ★

The diagram below shows a motor neurone.



Structure X of the neurone can be found in

A. the skin.
B. a spinal nerve.
C. the grey matter of the spinal cord.
D. the white matter of the cerebrum.

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0170000011|!|

Which of the following statements about axons is/are correct?

(1) In a motor neurone, the axon is usually longer than the dendrons.
(2) Axons transmit nerve impulses away from the cell body.
(3) Only axons are surrounded by myelin sheath.

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A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000012|!| ★

Interneurones are found in

A. the brain only.
B. the spinal cord only.
C. both the brain and the spinal cord.
D. neither the brain nor the spinal cord.

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000013|!|

Interneurones are found in

(1) the skin.
(2) the central nervous system.
(3) glands.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0170000014|!|

Which of the following types of neurones is/are connected to effectors?

(1) sensory neurones
(2) interneurones
(3) motor neurones

A. (1) only
B. (2) only
C. (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000015|!|

The table below shows the direction of nerve impulse transmission along three neurones, X, Y and X.

Neurone	From	To
X	sensory neurone	motor neurone
Y	central nervous system	effector
Z	receptor	central nervous system

What types of neurones are X, Y and Z?

	X	***Y***	***Z***
A.	sensory neurone	motor neurone	interneurone
B.	motor neurone	sensory neurone	interneurone
C.	interneurone	motor neurone	sensory neurone
D.	interneurone	sensory neurone	motor neurone

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0170000016|!|

Which of the following correctly identifies the types of neurones which transmit nerve impulses within the human body?

	Within the central nervous system	***From a receptor to the central nervous system***	***From the central nervous system to an effector***
A.	sensory neurone	motor neurone	interneurone
B.	motor neurone	sensory neurone	interneurone
C.	interneurone	motor neurone	sensory neurone
D.	interneurone	sensory neurone	motor neurone

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000017|!|

Which of the following comparisons between a sensory neurone and a motor neurone is correct?

	Sensory neurone	***Motor neurone***
A.	transmits nerve impulses away from the central nervous system	transmits nerve impulses towards the central nervous system
B.	the cell body lies outside the central nervous system	the cell body lies inside the central nervous system
C.	the dendron is short	the dendron is long
D.	the axon is long	the axon is short

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000018|!|

Which of the following statements about myelin sheath is/are correct?

(1) It is a fatty substance.
(2) It slows down the transmission of nerve impulses along a nerve fibre.
(3) Some nerve fibres are not surrounded by myelin sheath.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000019|!|

Myelinated axons make up the brain's

A. cavities.
B. meninges.
C. white matter.
D. grey matter.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0170000020|!| ★★ **STSE**

Multiple sclerosis is a disease in which the myelin sheath surrounding the axons of neurones is broken down. One of the symptoms of multiple sclerosis is weakening of muscles. Which of the following could be a possible cause for this weakening of muscles?

A. The transmission of nerve impulses along the axons of neurones is slowed down.
B. Synthesis of neurotransmitters is inhibited.
C. Receptors on the muscle fibres become insensitive to neurotransmitters.
D. Neurotransmitters are inhibited from travelling along the neurones.

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000021|!| ★

At the synapse between two neurones,

A. an electrical impulse is conducted through the solution in the synapse.
B. an electrical impulse jumps across the synapse cleft.
C. an electrical impulse from the first neurone stimulates the next neurone.
D. a chemical is released from the first neurone and diffuses to the next neurone.

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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000022|!| ★

Which of the following result in the transmission of nerve impulses in one direction only?

(1) Only the endings of an axon can secrete neurotransmitters.
(2) Only dendrons possess the receptor sites to bind with neurotransmitters.
(3) One neurone only receives neurotransmitters from another single neurone.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000023|!| ★ **Cross topic**

Neurotransmitters are released by

A. axon terminals.
B. dendrites.
C. the cell body.
D. motor end plates.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000024|!| ★ **Cross topic**

Which of the following can release neurotransmitters?

(1) axon of a sensory neurone
(2) dendron of an interneurone
(3) axon of a motor neurone

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A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000025|!| ★

Which of the following comparisons between the brain and the spinal cord is correct?

	Brain	***Spinal cord***
A.	protected by the cranium	protected by the vertebral column
B.	nourished by the cerebrospinal fluid	not nourished by the cerebrospinal fluid
C.	outer layer made up of cell bodies of neurones	outer layer made up of nerve fibres
D.	not involved in reflex actions	involved in reflex actions

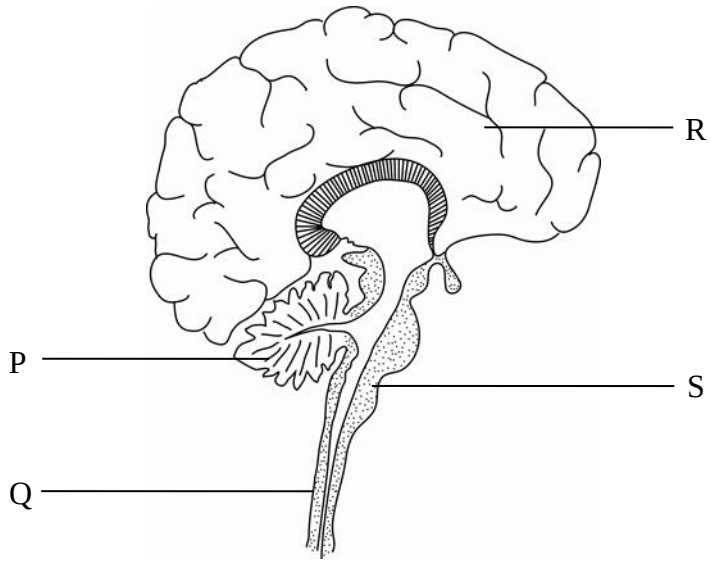
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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000026|!|

The diagram below shows a section of the human central nervous system.



Which parts of the central nervous system are responsible for the following functions?

	Body balance	***Breathing rate***	***Memory***
A.	P	Q	R
B.	R	S	P
C.	P	S	R
D.	S	Q	R

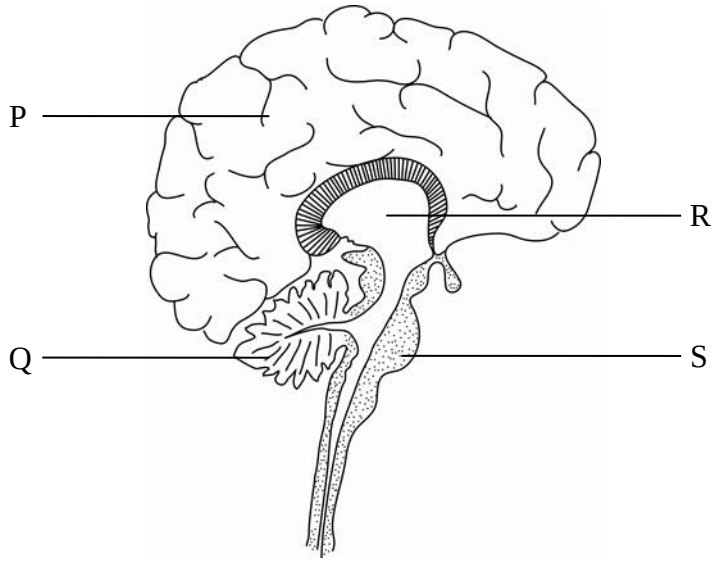
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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000027|!| ★

The diagram below shows a section of the human brain.



Which of the following statements about the brain is ***incorrect***?

A. Structure P is responsible for reasoning and judgement.
B. Structure Q coordinates the knee jerk reflex.
C. The fluid in structure R acts as a shock absorber.
D. Structure S coordinates swallowing.

□

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0170000028|!| ★

Alcohol affects muscle coordination by affecting the

A. cerebrum.
B. cerebellum.
C. hypothalamus.
D. spinal cord.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000029|!| ★

Which of the following is controlled by the cerebrum?

A. peristalsis
B. dilation of pupil
C. eye accommodation
D. knee jerk reflex

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000030|!| ★

Which of the following is/are controlled by the medulla oblongata?

(1) heart beat
(2) body balance
(3) peristalsis

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

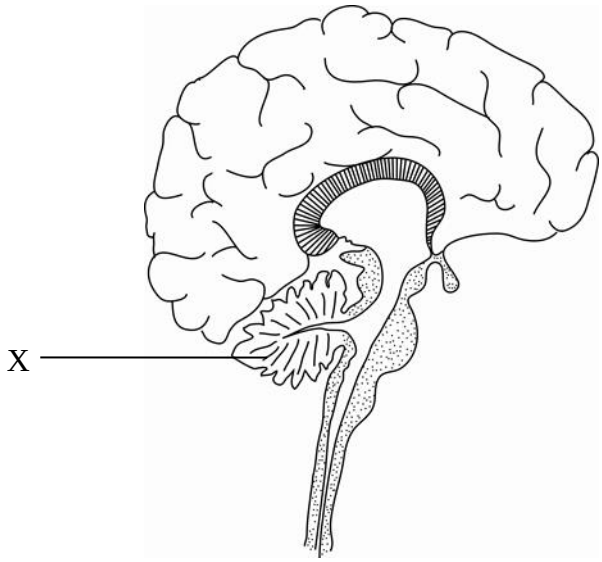
##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0170000031|!| ★

The diagram below shows a section of the human brain.



Which of the following is/are the possible symptom(s) of structure X being damaged?

(1) loss of memory
(2) slurred speech
(3) irregular heart beat

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000032|!| ★

Breathing movements are controlled by

(1) the cerebrum.
(2) the cerebellum.
(3) the medulla oblongata.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

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##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000033|!| ★

Which of the following parts of the brain is/are involved when a person is riding a bicycle?

(1) the cerebrum
(2) the cerebellum
(3) the medulla oblongata

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

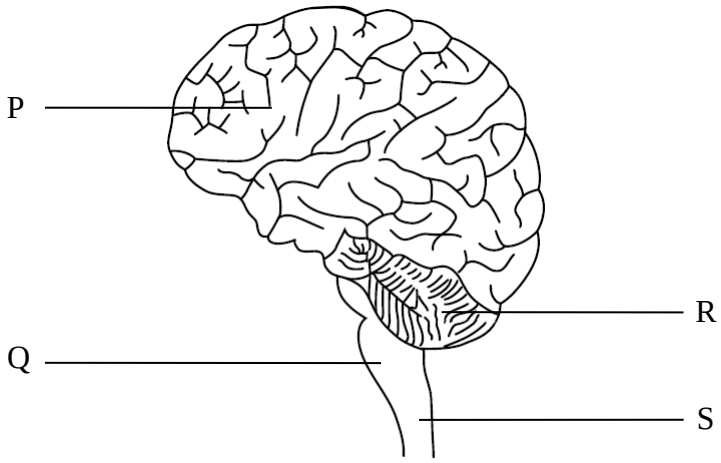
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##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000034|!| ★

The diagram below shows part of the human central nervous system.



Which of the following combinations correctly matches the structure and its function?

	Structure	***Function***
A.	P	regulating the muscle tone of the biceps
B.	Q	responsible for body balance
C.	R	controlling movement of the arm
D.	S	as a control centre of reflex actions

□

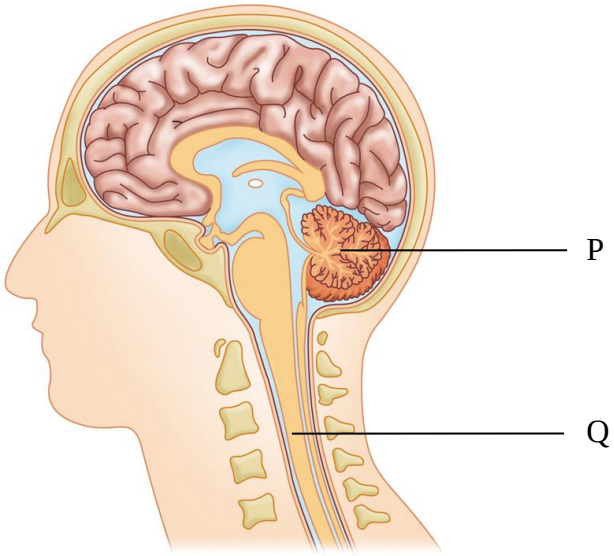
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##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000035|!| ★

The diagram below shows a section of the human head.



Which of the following statements about structures P and Q is correct?

A. Both have highly folded surfaces.
B. Both have grey matter inside and white matter outside.
C. Both are nourished by the cerebrospinal fluid.
D. Both are coordinating centres for simple reflexes.

☐

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000036|!| ★

Underdevelopment of the medulla oblongata would lead to

A. limited language skills.
B. poor processing of sensory information.
C. jerky uncoordinated movements.
D. irregular changes in blood pressure.

☐

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0170000037|!| ★

The cerebrospinal fluid contains

(1) oxygen.
(2) carbon dioxide.
(3) red blood cells.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000038|!|

Which of the following statements about the cerebrospinal fluid are correct?

(1) It fills the central canal of the spinal cord.
(2) It nourishes the central nervous system.
(3) It serves as a shock absorber.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

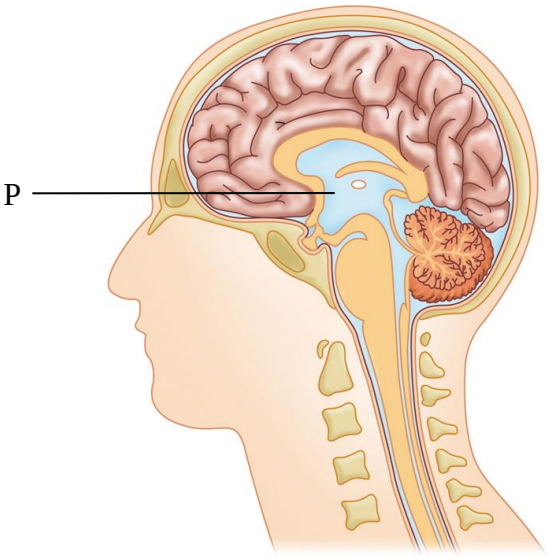
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0170000039|!|

The diagram below shows a section of the human head.



Which of the following are the functions of the fluid in space P?

(1) It acts as a lubricant.
(2) It supplies nutrients to the brain cells.
(3) It absorbs shock.

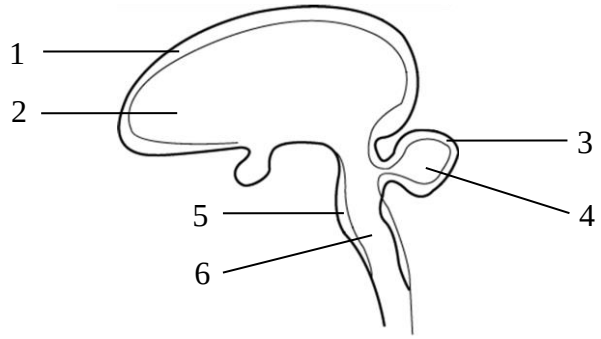
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000040|!|

The diagram below outlines a section of the brain.



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Which of the following shows the locations of the grey matter in the cerebrum, cerebellum and medulla oblongata of the brain?

	Cerebrum	***Cerebellum***	***Medulla oblongata***
A.	1	3	5
B.	2	4	5
C.	1	3	6
D.	2	4	6

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0170000041|!|

Which of the following combinations correctly identifies the distribution of white matter and grey matter in the cerebellum and the spinal cord?

	Inner part of the cerebellum	***Inner part of the spinal cord***
A.	white matter	grey matter
B.	white matter	white matter
C.	grey matter	grey matter
D.	grey matter	white matter

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000042|!| ★

Which of the following structures of the nervous system have the same arrangement of white matter and grey matter?

(1) cerebrum
(2) medulla oblongata
(3) spinal cord

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000043|!|

Which of the following statements about grey matter in the nervous system is ***incorrect***?

A. It contains the cell bodies of neurones.

B. It makes up the cerebral cortex.

C. It makes up the outer layer of the medulla oblongata.

D. It makes up the inner part of the spinal cord.

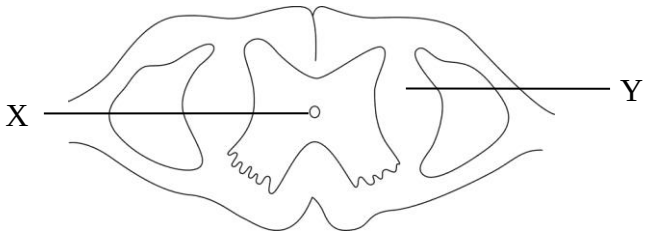
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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000044|!|

The diagram below shows the cross section of a human spinal cord.



Which of the following combinations correctly identifies X and Y?

	X	***Y***
A.	grey matter	white matter
B.	central canal	grey matter
C.	white matter	grey matter
D.	central canal	white matter

□

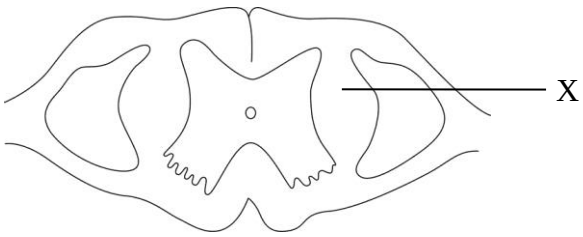
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0170000045|!| ★

The diagram below shows the cross section of a human spinal cord.



Region X of the spinal cord appears white due to the presence of a large amount of

A. cell bodies of neurones.
B. myelinated nerve fibres.
C. synapses.
D. neurotransmitters.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0170000046|!|

The dorsal roots of spinal nerves contain the fibres of

A. sensory neurones only.
B. motor neurones only.
C. interneurons only.
D. sensory neurones and motor neurones only.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0170000047|!| ★★

Damage to the surface tissue of the spinal cord would most likely affect

A. stimulation of sensory receptors.
B. communication between the peripheral nervous system and the brain.
C. information processing.
D. functioning of reflex arcs.

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##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0170000048|!|

Which of the following correctly describes reflex actions?

A. Reflex actions are automatic.
B. The brain always coordinates the responses.
C. All reflex actions involve the spinal cord.
D. All reflex actions are triggered by external stimuli.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000049|!| ★

Which of the following are correct about reflex actions?

(1) They are triggered by stimulation of receptors.
(2) Nerve impulses always pass from the receptor to the effector via the brain.
(3) The effector produces the response upon stimulation from the motor neurone.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000050|!|

Which of the following shows the nervous pathway that brings about the withdrawal reflex?

A. receptor → sensory neurone → interneurone → motor neurone → effector
B. receptor → sensory neurone → motor neurone → effector
C. receptor → sensory neurone → effector → interneurone → brain
D. receptor → motor neurone → effector → interneurone → brain

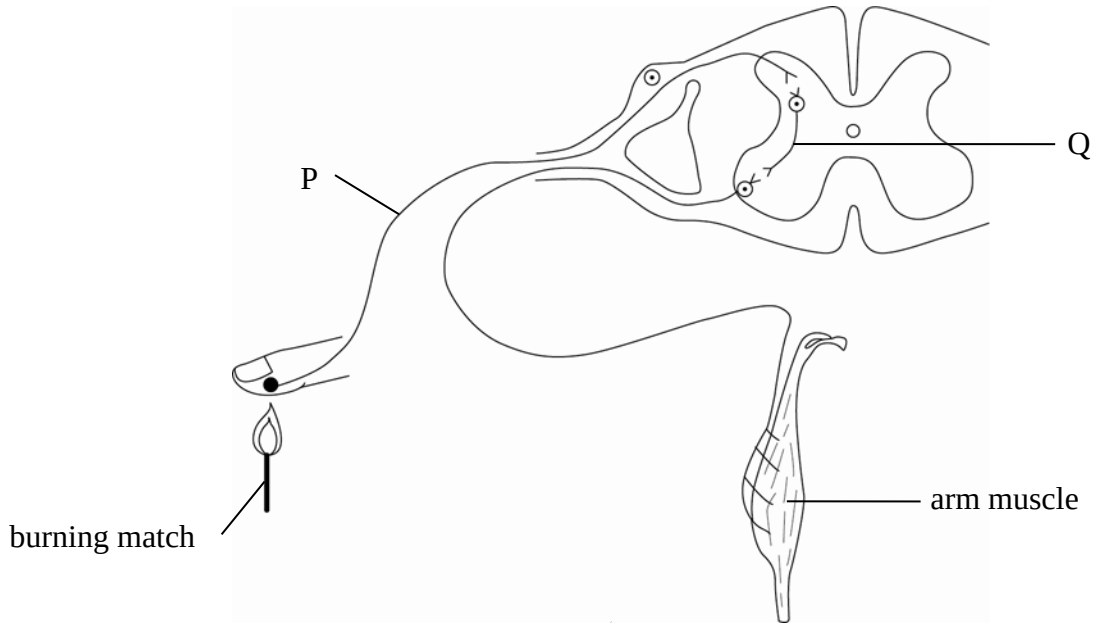
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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000051|!|

The diagram bellow shows the nervous pathway of a reflex action.



What types of neurones are P and Q?

	P	***Q***
A.	sensory neurone	interneurone
B.	sensory neurone	motor neurone
C.	motor neurone	interneurone
D.	motor neurone	sensory neurone

□

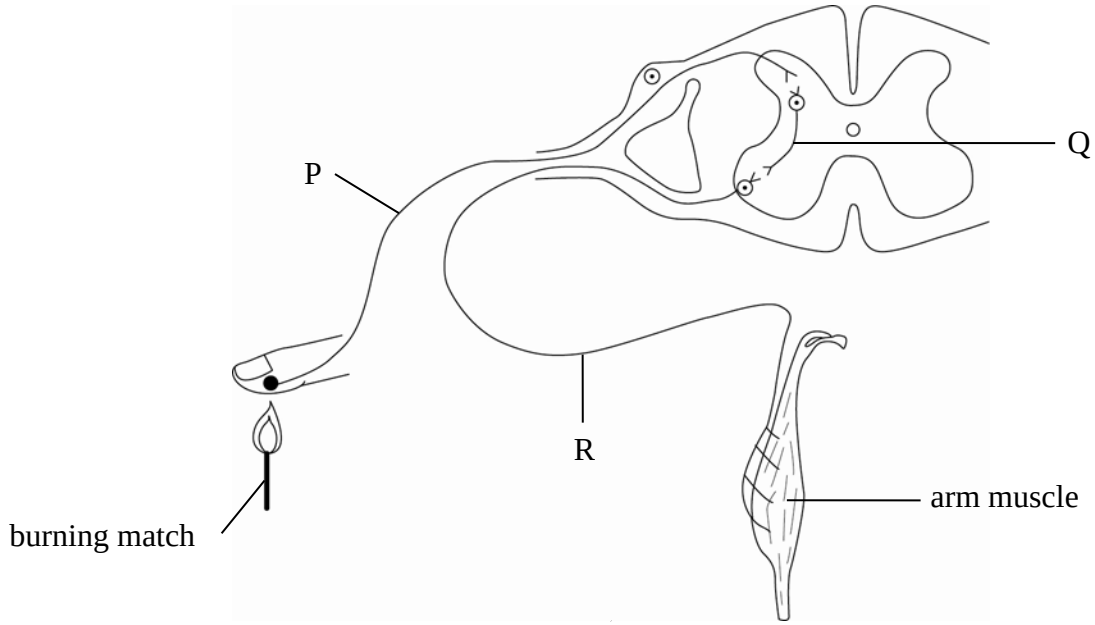
##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0170000052|!| ★ **Cross topic**

The diagram bellow shows the nervous pathway of a reflex action.



Which of the following correctly shows the nature of signal transmission between neurones P and Q, and that between neurone R and the arm muscle?

	Between neurones P and Q	***Between neurone R and the arm muscle***
A.	chemical	electrical impulse
B.	electrical impulse	electrical impulse
C.	chemical	chemical
D.	electrical impulse	chemical

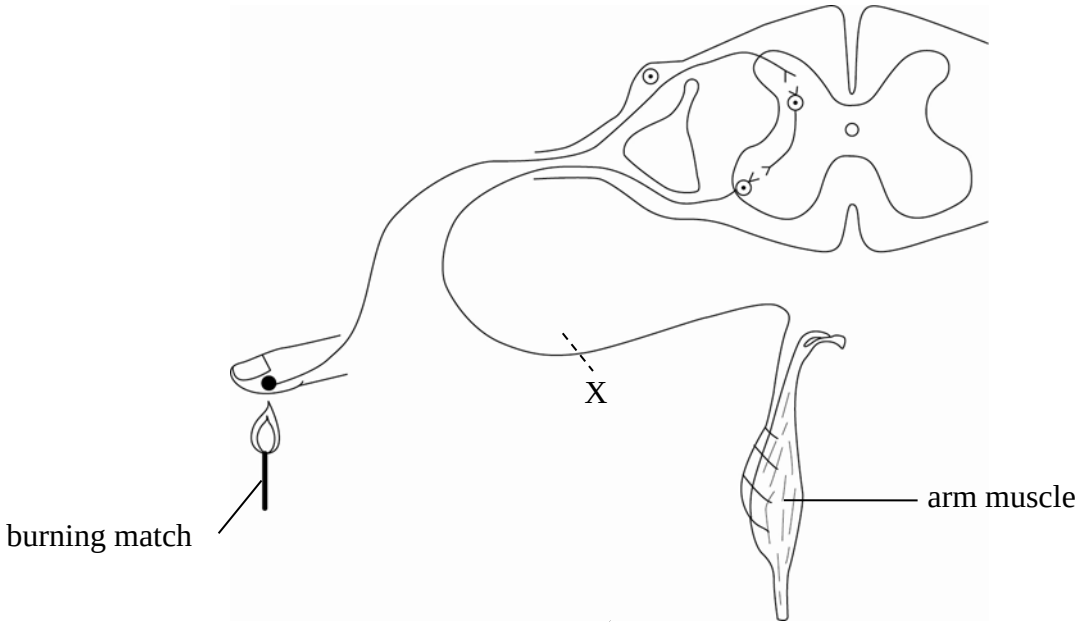
##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0170000053|!| ★

The diagram bellow shows the nervous pathway of a reflex action.



What would happen if a cut were made at X?

A. The person would feel the pain and withdraw his hand immediately.

B. The person would feel the pain but would not withdraw his hand immediately.

C. The person would withdraw his hand immediately but would not feel the pain.

D. The person would not withdraw his hand immediately and would not feel the pain.

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMA0170000055|!| ★★

Amy accidentally touched a hot pan. She withdrew her arm, and an instant later, she felt a burning pain. How would you explain the fact that her arm moved before she felt the pain?

A. The nervous pathway to the brain was blocked temporarily.
B. Her response was coordinated by the spinal cord, and it occurred before nerve impulses could reach the brain.
C. Nerve impulses travelled faster to her arm muscles because the motor neurones are myelinated.
D. This is not possible. The brain must be informed of the pain before a person can react.

□

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000056|!| ★

A man pricked his finger on a pin and withdrew his hand immediately. Which of the following statements about this reflex action is/are correct?

(1) The man withdrew his hand first, and then felt the pain.
(2) It is acquired through learning.
(3) It involves the central nervous system.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

□

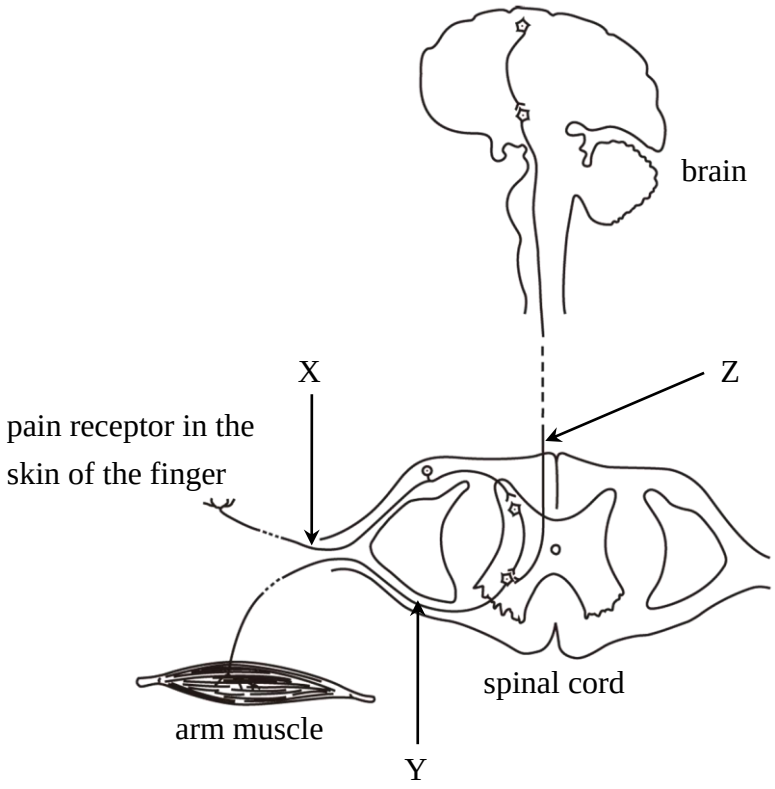
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0170000058|!| ★★ **STSE**

The diagram below shows the arrangement of some neurones in the nervous system.



An anaesthetic is a drug used to block nerve impulses. A person injected with an anaesthetic could move his arm in response to a pinprick but could not feel it. Where was the anaesthetic injected in this person?

A. at X
B. at Y
C. at Z
D. at X and at Z

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0170000059|!|

Which of the following shows the nervous pathway that brings about the knee jerk reflex?

A. receptor → sensory neurone → interneurone → motor neurone → effector
B. receptor → sensory neurone → motor neurone → effector
C. receptor → sensory neurone → effector → interneurone → brain
D. receptor → motor neurone → effector → interneurone → brain

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000060|!|

In a medical test, the doctor gently tapped the hanging leg of a patient just below the knee cap. Which of the following best describes the response of the patient?

A. He felt the tapping action and his hanging leg kicked forwards at the same time.
B. He felt the tapping action, and then his hanging leg kicked forwards.
C. His hanging leg kicked forwards, and then he felt the tapping action.
D. His hanging leg kicked forwards but he would not feel the tapping action.

##C##

Skill assessed	Mark
Understanding basic concepts	1

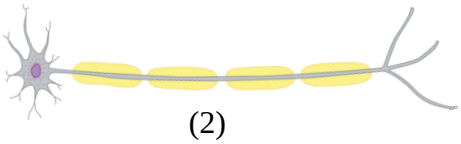
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|!|BCEMB0170000062|!| ★

The diagram below shows three types of neurones.



(1)



(2)



(3)

The nervous pathway of the knee jerk reflex involves

A. (1) and (2) only.
B. (1) and (3) only.
C. (2) and (3) only.
D. (1), (2) and (3).

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000063|!| ★

Which of the following about the knee jerk reflex is ***incorrect***?

A. The stimulus is detected by the stretch receptor in the upper thigh muscle.
B. Interneurones are not involved in the knee jerk reflex.
C. The brain acts as the coordination centre.
D. The upper thigh muscle contracts when it receives nerve impulses from the motor neurone.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0170000064|!| ★

Which of the following comparisons between the nervous coordination of kicking a football and the knee-jerk reflex is ***incorrect***?

	Kicking a football	***Knee jerk reflex***
A.	under conscious control	not under conscious control
B.	slower response	faster response
C.	does not involve the spinal cord	involves the spinal cord
D.	requires learning	does not require learning

□

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0170000065|!|

Which of the following comparisons between a reflex action and a voluntary action is/are correct?

(1) A reflex action is inborn while a voluntary action is not inborn.
(2) A reflex action does not involve the cerebrum while a voluntary action involves the cerebrum.
(3) The speed of response of a reflex action is always faster than that of a voluntary action.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0170000066|!| ★

Which of the following comparisons between a reflex action and a voluntary action is ***incorrect***?

	Reflex action	***Voluntary action***
A.	does not involve the brain	involves the brain
B.	no learning is needed	learning is needed
C.	involves receptors	may not involves receptors
D.	same stimulus always results in the same response	responses may vary

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000067|!| ★

Which of the following is ***not*** a reflex action?

A. Blinking of the eyes when foreign particles get into the eyes.
B. Sneezing of the nose when the nasal passages detect irritating particles in the air.
C. Withdrawal of the arm when a person pricks his finger on a pin accidentally.
D. An athlete starts to sprint as soon as soon as he hears the starter's signal.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000068|!| ★

Which of the following is a voluntary action?

A. answering the door when the doorbell rings
B. dilation of the pupil in dim light
C. increase in the breathing rate during vigorous exercise
D. increase in salivation when chewing

##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0170000069|!| ★

Which of the following does ***not*** involve the brain as the coordinating centre?

A. swallowing
B. constriction of the pupil under bright light
C. kicking of the leg when the tendon below the knee cap is tapped
D. crying of a baby when it is hungry

□

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0170000070|!| ★

A person heard the telephone rang and then picked up the receiver. Which of the following shows the nervous pathway involved?

A. sensory hair cells in the cochlea → spinal cord → auditory nerve → cerebrum → motor neurone → arm muscle
B. sensory hair cells in the cochlea → auditory nerve → spinal cord → cerebrum → motor neurone → arm muscle
C. sensory hair cells in the cochlea → auditory nerve → cerebrum → spinal cord → motor neurone → arm muscle
D. sensory hair cells in the cochlea → auditory nerve → cerebellum → spinal cord → motor neurone → arm muscle

□

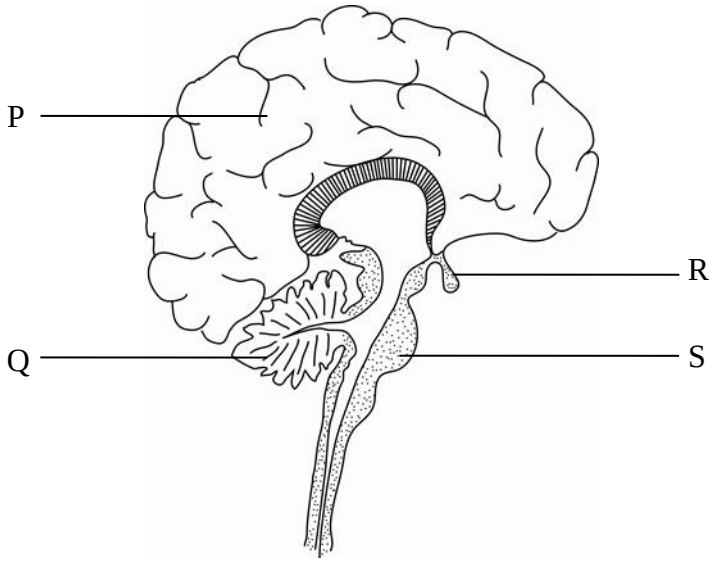
##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0170000071|!|

The diagram below shows a section of the human brain.



Which structure of the brain secretes hormones?

A. P

B. Q

C. R

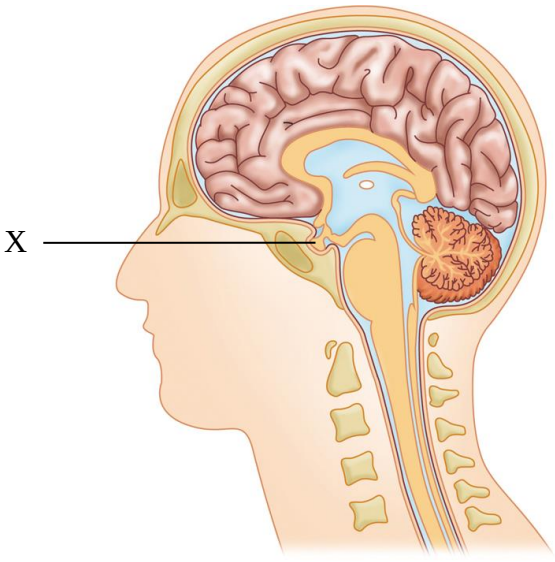
D. S

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000072|!| ★

The diagram below shows a section of the human head.



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Which of the following hormone(s) is/are secreted from structure X?

(1) growth hormone
(2) thyroxine
(3) progesterone

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000073|!| ★
Which of the following is/are exocrine gland(s)?

(1) salivary gland
(2) pituitary gland
(3) pancreas

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000074|!| ★
Which of the following are endocrine glands?

(1) pituitary gland
(2) gastric gland
(3) ovary

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A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000075|!| ★ **Cross topic**

Which of the following statements about hormones is correct?

A. Hormones are released through ducts.
B. Hormones are released by ductless glands.
C. Each hormone affects only one target organs.
D. They are always regulated by negative feedback mechanism.

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0170000076|!| ★ **Cross topic**

Which of the following correctly matches the hormone and its function?

	Hormone	***Function***
A.	adrenaline	decreases heart rate
B.	anti-diuretic hormone	promotes reabsorption of water in kidneys
C.	glucagon	stimulates conversion of glucose to glycogen
D.	Follicle stimulating hormone	stimulates ovulation

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0170000077|!|

Hormonal coordination differs from nervous coordination in that

A. the brain is not involved in hormonal coordination.
B. the transmission of messages is faster.
C. the effect of hormone coordination is more localized.
D. the effect of hormone coordination lasts longer.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0170000078|!|

Which of the following comparisons between nervous coordination and hormonal coordination is ***incorrect***?

	Nervous coordination	***Hormonal coordination***
A.	the action is faster	the action is slower
B.	the response is short-term	the response is long-lasting
C.	the effect is widespread	the effect is localized
D.	messages are electrochemical in nature	messages are chemical in nature

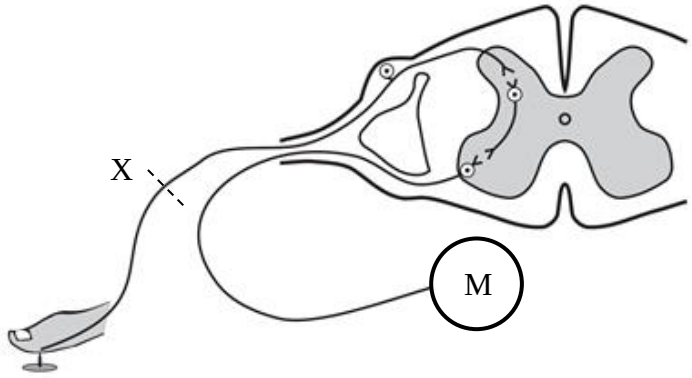
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0170000079|!| ★

Directions: The following two questions refer the diagram below which shows the nervous pathway of a reflex action that occurs when a finger touches a sharp pin.



What is structure M?

A. motor neurone
B. muscle
C. pain receptor
D. the brain

☐

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000080|!| ★

What would happen if a cut were made at X?

A. The person would feel the pain and withdraw his hand immediately.
B. The person would feel the pain but would not withdraw his hand immediately.
C. The person would withdraw his hand immediately but would not feel the pain.
D. The person would not withdraw his hand immediately and would not feel the pain.

☐

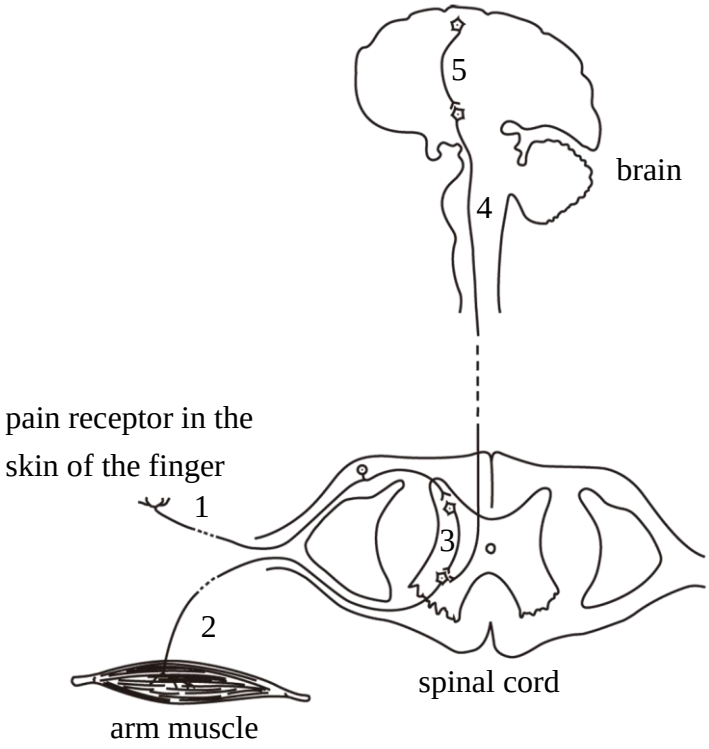
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0170000081|!| ★

Directions: The following two questions refer the diagram below which shows the arrangement of some neurones in the nervous system.



Which of the following correctly shows the nervous pathway for the withdrawal reflex of the arm?

A. 1 → 3 → 2
B. 1 → 3 → 4 → 5
C. 5 → 4 → 2
D. 5 → 4 → 3 → 1

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000082|!| ★

Which of the following correctly shows the nervous pathway for the voluntary movement of the arm?

A. 1 → 3 → 2
B. 1 → 3 → 4 → 5
C. 5 → 4 → 2
D. 5 → 4 → 3 → 1

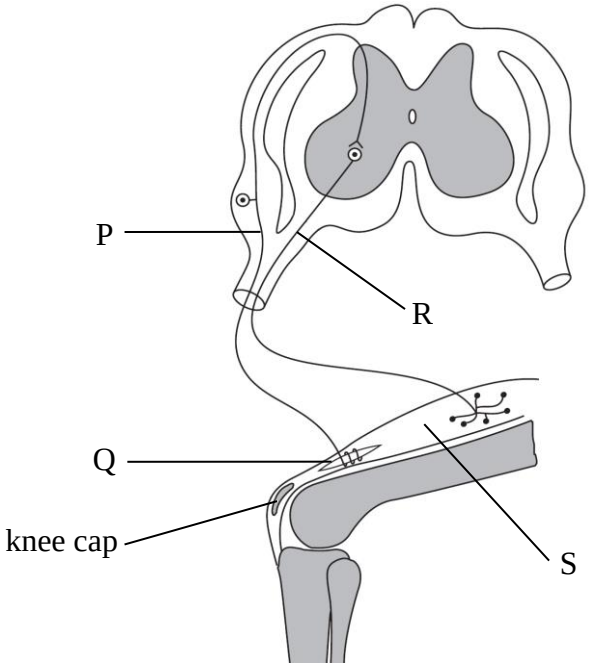
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##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000083|!| ★

Directions: The following two questions refer the diagram below which shows the nervous pathway of the knee jerk reflex. When the person is tapped gently below the knee cap, the lower leg kicks forwards.



The motor neurone is labelled

A. P.
B. Q.
C. R.
D. S.

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0170000084|!| ★

Another nervous pathway allows the person to feel the tapping action. This pathway connects to

A. the cerebellum.
B. the cerebral cortex.
C. the spinal cord.
D. the medulla oblongata.

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##B##

Skill assessed	Mark
Understanding basic concepts	1

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